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Free Peanuts and Expensive Batteries

Supply and Demand in Action

The story is told of two economists on their way to lunch when they spot what appears to be a £50 note lying on the pavement. When the younger economist bends to pick it up, his older colleague restrains him, saying, 'That can't be a £50 note.' 'Why not?' asks the younger colleague.

'If it were,' comes the reply, 'someone would have picked it up by now.'

The older economist may have been wrong, of course. Yet his admonition embodies an important truth that people often ignore. The 'no cash on the table' principle holds that freely available money seldom sits unclaimed for long. In the future, as in the past, the only way to make real money will be through some combination of talent, thrift, hard work and luck.

Millions of people, however, seem to believe they can get rich in a hurry. They saw others do it in the 1990s simply by transferring their money from old-economy stocks, such as General Electric or Procter & Gamble, into Oracle, Cisco Systems and

other high-tech stocks that led the explosive rise of the NASDAQ Indices. More recently they saw others get rich overnight by borrowing as much as they could and buying property that, by traditional standards, was well beyond their reach.

People who think they can spot cash on the table offer confident explanations for why the usual constraints do not apply. Many bullish stock analysts in the 1990s, for example, insisted that traditional valuation formulas were no longer valid because the Internet was changing the rules of the game. And with business-to-business e-commerce reducing some companies' operating costs by more than 30 per cent, there was little doubt that new technologies were creating spectacular productivity gains.

But as everyone now realises, and should have realised even then, the ultimate value of an e-commerce company depends not on the productivity gains it makes possible but on how much profit it generates. New technologies will continue to generate a burst of profits for companies that are relatively quick to adopt them. But, as in the past, when competitors adopt the same technologies, the long-run savings will be captured not by producers in the form of higher profits but by consumers in the form of lower prices. Thus US dairy farmers who were quick to adopt bovine somatotropin, the hormone that increases milk yields by as much as 20 per cent, reaped a short-term windfall. But as use of the hormone spread, increased production pushed milk prices steadily downward, eroding profit margins.

A similar profit trajectory ultimately characterised most purveyors of new technologies. Organisers of business-to-business e-commerce may indeed have saved manufacturers billions of pounds. But because the new technology companies are no more insulated from competition than dairy farmers, most of those savings took the form of lower product prices, not higher profits. The 'no cash on the table' principle reminds us to be wary of opportunities that seem too good to be true. It predicted the

spectacular US NASDAQ crash and the corresponding UK FTSE falls that occurred in March 2000. But together with the cost-benefit principle, it also helps us understand less spectacular patterns in ordinary markets. Consider, for example, the prices at which products are sold.

Because people's tastes and incomes differ, the amount they are willing to pay for any good typically varies across a broad range. But as Adam Smith argued in *The Wealth of Nations*, the price of a product should not exceed the cost of producing it in the long run. Otherwise, profit opportunities would induce rival sellers to enter the market. And entry would continue until added supply drove price down to cost.

Yet examples abound in which different buyers appear to pay substantially different prices for essentially identical goods and services. These examples seem to contradict the 'no cash on the table' principle. Why doesn't competition from rival sellers force all prices to the same level? Many of the examples we'll take up in Chapter 4 bear directly on this question. For now, suffice to say that in many markets, competition does drive price to a common level.

Within very narrow limits, for example, gold sells for the same price in London as in New York, and for the same price to company executives and primary school teachers. If that were not the case, there would be cash on the table. Suppose, for example, that an ounce of gold sold for £400 in New York and £450 in London. Someone could then buy an ounce of gold in New York and immediately turn a profit of £50 by reselling it in London. The law of one price – which is really just a restatement of the 'no cash on the table' principle – says that the price of gold in the two cities generally will not differ by more than the cost of shipping the stuff from one to the other.

The law of one price applies most forcefully to commodities and services that are sold in perfectly competitive markets.

Roughly speaking, these are markets in which numerous suppliers sell highly standardised products. The gold market is a classic example. Gold is a highly standardised commodity, and it is relatively easy for new firms to enter the market whenever and wherever profit opportunities arise.

It is the possibility of arbitrage – purchase at one price with a riskless opportunity to resell at a higher price – that enforces the law of one price. A rich man might be willing to pay more than a poor man for a kilogram of standard table salt, if only because he has greater ability to pay. Yet the price of salt is the same for all. The law of one price says that any supplier who tried to exploit the rich man's willingness to pay more would create immediate profit opportunities for rival sellers. And even if sellers conspired to maintain higher prices for rich buyers, poor arbitrageurs could symmetrize their efforts. They could buy salt at the poor man's price and resell it to the rich at a profit by charging just a little less than the rich man's price. As more and more poor men tried to get a piece of this action, the price difference would be driven closer and closer to zero.

The economist's model of supply and demand is essentially a story about the forces that determine which products get produced in which quantities and at what prices. The demand for a given product is a measure of how much people are willing to pay for it. It is a summary, in other words, of the benefits they feel they receive from consuming it. People will keep buying more of a product as long as the value they assign to the last unit consumed is at least as great as its price. The general pattern is that as the price of a good keeps rising, the quantity demanded keeps falling.

The supply of given product is a summary measure of the terms on which producers are willing to offer it for sale. The basic supply rule is that producers will keep offering products for sale as long as the price for which they can sell it is at least as

great as their marginal cost — the cost of producing the last unit supplied. In the short run, marginal cost tends to rise with the number of units produced. (This is in part a consequence of the 'low-hanging fruit' principle, which says that it is always best to exploit one's best opportunities first.) So on the supply side, the general pattern is that as the price of a good keeps rising, sellers are willing to sell more units.

The market for any given product is said to be in equilibrium when the amount consumers wish to buy at the prevailing price is the same as the amount producers wish to sell. The equilibrium price is also called the *market-clearing price*.

The supply and demand model has enormous power to extract orderly patterns from the cacophony of information that bombards us each day in the marketplace.

Because the market price results when the supply side and the demand side of the market are in balance, it is never correct, strictly speaking, to explain price or quantity movements by referring only to supply or only to demand. Nevertheless, many important patterns in the marketplace can be understood by focusing on sellers in some cases and on buyers in others. The phenomena described in the first sequence of examples are largely driven by the demand (or buyer's) side of the transaction.

Why do many bars charge customers for water but give them peanuts for free?

Some bars charge their patrons as much as £2 for a half-litre bottle of water, while making sure that full bowls of free salted nuts are always within easy reach. Since nuts are more costly to produce than water, shouldn't it be the other way around?

The key to understanding this practice is to recognise that the terms on which bars offer both water and nuts are dictated by the effect of these commodities on demand for bars' core prod-

uct, alcoholic beverages. Nuts and alcoholic beverages are complements. Someone who eats more nuts will demand more beer or spirits. Since nuts are relatively cheap and each alcoholic drink generates a relatively high profit margin, making nuts freely available tends to increase bars' profits.

By contrast, water and alcoholic beverages are substitutes. The more water bar customers drink, the fewer alcoholic beverages they will order. So even though water is relatively inexpensive, bars have an incentive to set a high price for it, thereby discouraging its consumption.

Why do many computer manufacturers offer free software whose market value exceeds the price of the computer itself?

Someone who purchases a new computer today will find that its hard drive already contains not only the latest operating system but also the latest word processing, spreadsheet, presentation, e-mail, music and photo software, not to mention the latest in virus protection. Why give away all this valuable software?

Software users care a great deal about product compatibility. When scientists or historians are working on a project together, for example, their task will be much simpler if they all use the same word processing program. Likewise, a self-employed person's life will be easier at tax return time if her financial software is the same as her accountant's.

A related consideration is that many programs, such as Microsoft Word, are challenging to master. People who become proficient in a program often resist learning another, even if it is objectively better.

The implication is that the benefit of owning and using any given software program increases with the number of other people who use it. This unusual relationship gives the producers of

the most popular programs an enormous advantage and makes it hard for new programs to break into the market.

Recognising this pattern, Symantec Corporation offered computer makers free trial copies of their Norton anti-virus software. Computer makers were only too happy to include the program, since it made their new computers more attractive to buyers. By giving away free copies of the program, Symantec 'primed the pump', creating an enormous demand for their products, including upgrades of Norton Anti Virus and more advanced versions of related software.

Inspired by this success story, other software developers have jumped onto the bandwagon. Some software developers are even rumoured to *pay* computer makers to include their programs.

Why does a mobile phone sell for only £12.95, while a spare battery for that same phone sells for £19.99?

(Tianxin Gu)

At the Carphone Warehouse you can get the Nokia 6300 mobile phone 'free' with a twelve-month contract. But if you buy an additional battery for your mobile phone (perhaps because you expect to be away from your charger for extended periods), you'll pay £19.99. Why must you pay so much more for a replacement battery that is exactly the same as the one in your cheaper mobile phone?

Rechargeable lithium ion batteries of the sort used in mobile phones are costly to produce. So perhaps the more interesting question is why the original mobile phone/battery combination was so cheap. The answer seems to lie in the distinctive cost structure of wireless communication providers. Most of the costs incurred by these companies are fixed costs associated with setting up their networks – constructing mobile-phone masts, acquiring relevant licences and so on. These costs, as well as advertising

expenditures, do not vary with the amount of service they provide. The more customers a wireless service provider can attract to help defray these costs, the more likely it is to remain in business.

Suppose the monthly fee for a representative service contract is £30. Any company that manages to sign up one additional customer will then enjoy £360 a year in additional revenue, yet will incur no significant additional costs. Wireless service providers thus face powerful incentives to attract additional customers.

Mobile phones and wireless services are strong complements. Experience has demonstrated that offering heavily discounted mobile phones is an effective tactic for attracting new customers. Because they buy in bulk, wireless service providers can negotiate heavy discounts with Nokia, Motorola and other mobile phone manufacturers. Many companies offer new customers phones for less than they themselves paid for them, and some provide 'free' phones to new customers. But if giving away a phone attracts an additional customer who pays £360 a year in subscription fees, that is a good deal for the wireless company even if it pays Motorola £50 for the phone.

In contrast, offering a discount on spare batteries has not emerged as a successful tactic for attracting new customers. (That is hardly surprising, since most people are not in the market for a spare battery most of the time.) Wireless companies thus find it advantageous to offer mobile phones for less than the price of the batteries that power them.

Why are the most expensive flats in a high-rise building in India those on the upper floors, while the most expensive ones in a low-rise building are those on the lower floors?

(Pankaj Badlani)

In high-rise apartment buildings in Bombay, the monthly rent for a flat on a given floor is 1 to 3 per cent greater than for a similar flat

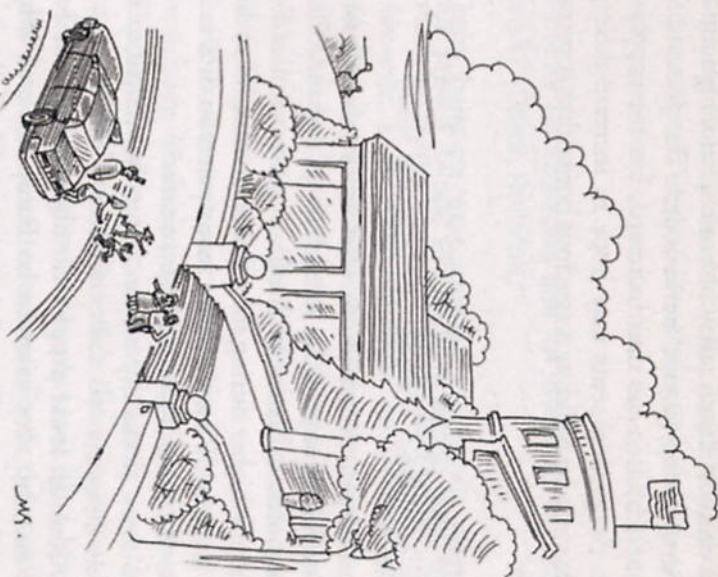
one floor down. A flat on the twentieth floor of a high-rise building might thus be let for 15 to 45 per cent more than it would if it were on the fifth floor. In apartment buildings with four storeys or fewer, however, the pattern is the reverse. Units on the first and second floors of such buildings can be let for significantly more than comparable ones on the third and fourth floors. Why this reversal?

In all cases, a higher apartment offers a better view and reduced street noise. These advantages are clearly greater for flats at the top of a high-rise building than for their low-rise counterparts. But even so, it would seem that the advantage of being higher up should command at least some rent premium, irrespective of total building height.

But there is a special twist in India – apartment buildings with four or fewer storeys are exempt from the law requiring apartment buildings to have lifts. In low-rise buildings, occupants of apartments on higher floors must thus haul their groceries up several flights of steps. And since the upper-storey flats don't offer dramatic views or even much insulation from street noise, most occupants would choose apartments on the lower floors of low-rise buildings if all rents were the same. The higher demand for such flats explains why they command higher rents.

Why do many people buy larger houses when they retire and their children leave home? (*Tobin Schilke*)

Many retired people continue living in the house where they brought up their family and then move to sheltered housing when they can no longer manage on their own. In past decades, when people tended to move after retirement, they typically bought smaller dwellings in seashore resorts, or some other location with a mild climate. Many still do, of course. But a recent trend is for retirees to abandon their existing homes in favour of much larger houses close by. Why are they doing this?



Large houses for retirees: grandchildren magnets?
DRAWING BY MICK STEVENS

One possibility is that current retirees are wealthier and can afford to move into larger houses. But why do they want larger houses after their children leave, and why do they build them close by? After all, they could afford to build or buy a larger dwelling somewhere on the south coast. Why build a large house in Cheshire?

A plausible conjecture is that a large house close to grown-up children's homes may lure the grandchildren to visit more often. With divorce and remarriage more common than in past

decades, many children today have six or more grandparents, if the parents of step-parents are included. The demand for visits from grandchildren has thus increased, but the supply of visits has not. So grandparents may hope to increase their share of available visits by building a spacious house that is conveniently located.

Why are hotel prices in Sharm El Sheikh lowest during periods of highest occupancy? (*Rhonda Hadi*)

The normal pattern is for hotel rates to vary directly with occupancy, which in turn varies directly with demand. Occupancy rates for hotels in Sharm El Sheikh, a resort town in Egypt, are substantially higher during the summer months than during the winter months. Why, then, are room prices in Sharm El Sheikh significantly lower in the summer months?

Hotel room prices depend on not only occupancy rates but also the willingness and ability of potential occupants to pay for them. Although fewer people visit Sharm El Sheikh during the winter months, they tend to be Europeans and other high-income Westerners. They choose Sharm El Sheikh because its weather promises a respite from chilly northern climates.

By contrast, tourists from Egypt and elsewhere in the Middle East do not face harsh winters and thus prefer to visit in the summer months, when school and work holidays are concentrated. Because these visitors generally have lower incomes than those who visit during the winter, hotels are unable to charge the high rates they command in winter.

THE PRECEDING EXAMPLES were explained primarily by differences on the demand side of the market. In each case, the focus was on why buyers might be willing to pay more for one product than another. The examples that follow describe phenomena

whose explanation lies primarily on the supply side. In each case, the unexpected price or product offering is linked in some way to differences in cost.

Why do colour photographs sell for less than black-and-white ones? (*Othron Roitman*)

When baby boomers were growing up, colour photographs often cost twice as much or more as standard black-and-white. Today, however, it is black-and-white photos that are more expensive. One high street photo chain, for example, charges from £9.99 to develop and print a 36-exposure roll of black-and-white film but only around £5.99 for the same size roll of colour film. Why this reversal?

In the 1950s, the consumer market for colour photography was in its infancy. Available processes for producing prints from colour film were much more complex and costly than the corresponding processes for black-and-white film. Because of this initial cost difference, most people shot photos in black-and-white, which gave shops an incentive to specialise in that medium. As volume grew, the resulting efficiency gains from specialisation further reduced the cost of processing black-and-white photos.

As long as black-and-white continued to be the dominant medium, colour processing remained an intrinsically more complex task. But as rising income led more consumers to opt for colour, manufacturers developed new optical machines that developed colour film and made prints automatically. These machines, which could cost as much as £70,000 apiece, were economical only if a store developed and printed a large volume of photos each day. Their compelling advantage was that they produced a large number of photos with little labour cost. And since labour costs had been the most important component of photo processing costs, stores with the new machines could

produce and sell colour prints far more cheaply than black-and-white ones.

Why couldn't the same automated machines produce black-and-white photos? Actually they could, but doing so required expensive paper, and the quality of the resulting photos was lower than those processed manually. So over the years, black-and-white photography increasingly became a niche market for professionals and serious amateurs.

Increasingly the norm is to use digital, as opposed to optical, processing machines. Because these machines can print black-and-white photos on the same paper used for colour photos, the costs of producing the two types will soon be essentially the same. Once this happens, the premium price for black-and-white photos should disappear.

Why do new cars costing £8,000 rent for £25 a day, while morning suits costing only £250 rent for around £45?

(John Gotte)

National car rental chains buy new cars in high volumes and thus are able to negotiate large discounts from manufacturers. They typically own a car for two years and then sell it for about 75 per cent of what they paid for it. So their opportunity cost of owning each car is much lower than a private consumer's would be.

By contrast, most morning suit rental shops are locally owned and operated. A medium-sized location will typically have around a thousand suits for hire, and its annual replacement purchases are not large enough to command large discounts in purchase prices. Because there is little resale market for used suits, they are often donated or sold off cheaply to school drama departments and orchestras. So whereas the fees charged by car rental companies must cover about a quarter of the purchase price of each car over

a two-year period, morning suit shops must charge fees that are sufficient to cover the full purchase price of each suit.

More importantly, the stock held by car rental companies tends to be more fully utilised than the stock held by morning suit shops. Most morning suit rentals are for events that fall on Saturdays. A shop with a thousand-suit stock might rent out one hundred of its suits on any given Saturday, but on other days of the week consider itself fortunate to rent out even five suits. By contrast, a substantial proportion of a car rental company's fleet goes out every day.

Another factor is that car rental companies often charge significantly more than their advertised rates by charging premium prices for add-ons. Insurance surcharges, for example, far exceed the cost of providing insurance, and customers who neglect to fill their petrol tanks are charged much more per litre than normal street prices.

Finally, a morning suit shop often has to alter a suit to fit the customer and may incur tailoring costs that are almost as high as the rental fee itself. Each suit must be dry-cleaned before it can be rented out again, which can add as much as £5 of additional expense for each rental. By contrast, a car rental company need only hose down a returned car before it is ready to go out again.

So even though the retail purchase price of a car is as much as forty times as high as the retail purchase price of a morning suit, it is not so surprising that the daily hire charge for a car is less than half the fee for a suit.

Why do many cleaners charge more for women's shirts than for men's? (Don Aday)

A small-town cleaners charges £3 for washing and pressing a woman's cotton shirt, but only £2 for a man's. Is it discriminating against women?

There is some evidence that women tend to pay more than men for expensive products like cars, whose selling price is usually negotiable. But laundry services do not fall in this category. Laundries typically post different prices for men's and women's garments, and customers almost never attempt to negotiate discounts from these prices.

In general, the more competitive an industry is, the less likely it is to subject customers to discrimination. Even a fairly small town will have more than a dozen laundry services listed in the Yellow Pages, which should be more than enough to ensure stiff competition. If existing laundries were charging prices significantly above their cost of processing women's shirts, there would be cash on the table. A rival firm could simply put up a sign saying, 'No additional charge for women's shirts' and quickly capture most of the women's market.

The persistence of price differentials suggests that they are rooted in differing costs of processing the two types of shirts. As in most service industries, the lion's share of costs incurred by laundries are labour costs. It is hard to imagine how it could be more costly to wash a woman's shirt than a man's. After all, shirts of both types go into the machine and are washed without further handling. So if there is a difference in cost, it must be in pressing the garments. Whenever possible, laundry workers iron shirts on a standard press, which greatly speeds the process. Shirts cannot be pressed on the ironing machine if they are too small or have delicate buttons or detailing. The standard press also clamps the shirt at the bottom, which leaves a conspicuous indentation on the garment. Shirts that cannot be done on a standard press must be done by hand, which takes much longer. In general the pressing machine accommodates men's shirts better than women's, which are often more delicate and thus more likely to be damaged by the machines. And because women commonly do not tuck their shirts into their trousers or

skirts, the large indentation left by the pressing machine on the bottom front of the garment is generally considered unacceptable. Men (at least until recently) tend to tuck their shirts in, so the indentation is not problematic for them.

In short, the most plausible explanation for why laundries charge more for women's shirts is that they are more costly, on average, to iron.

Why have Hindi-language movies been attracting much larger audiences in recent years? (Chris Anderson)

Until recently, someone from New Delhi living in the UK had to return to India to see movies in his native tongue. But it is now possible even for someone living in Rochdale, England to choose from hundreds of Hindi-language movie titles. What led to this change?

As Chris Anderson describes in his book *The Long Tail*, traditionally only residents of large cities have had access to foreign-language film screenings. Cinema owners cannot profitably book a film unless scores of paying customers attend each showing. That is a tall order for a Hindi film, even in cities with large populations of Indian immigrants.

With the advent of online DVD distribution services like Blockbuster, however, the market for relatively obscure movies has been radically transformed. To make money from such movies, it is no longer necessary to attract a significant number of viewers to the same location at the same time. If you want to see *Gol Mol*, a 1979 Hindi comedy featuring Palekar, a sports-obsessed job seeker whose new boss (Dutt) is a strict disciplinarian who forbids bringing up non-work related topics in the office, you need only add it to your Blockbuster queue. There is no city in the UK with an Indian population large enough to support cinema screenings of *Gol Mol*. But there is a large

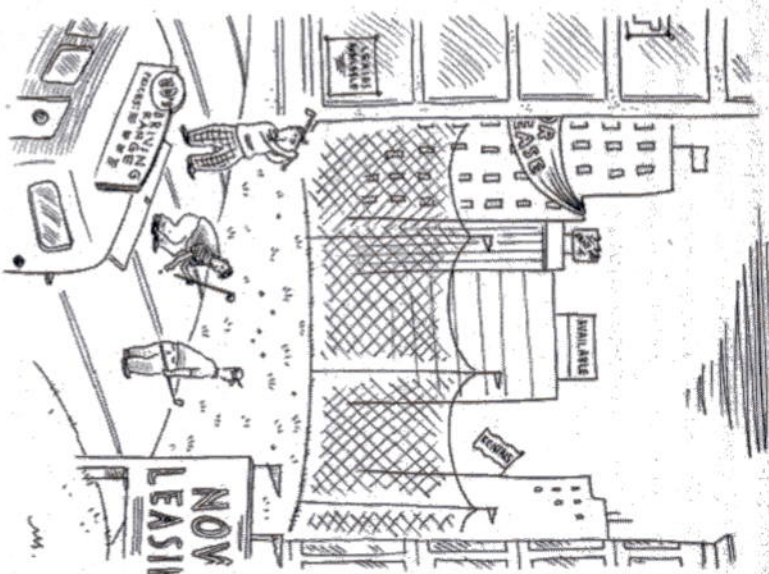
enough audience for such films to compensate Blockbuster for the small cost of adding them to its stocklist.

Hundreds of thousands of movies and books are insufficiently popular to justify film screenings or space in a commercial bookshop. The advent of online distribution has rescued these works from extinction.

Why did golf driving ranges spring up all over the suburbs of Washington, DC in the early 1990s? (Charles Kehler)

With trade associations and lobbyists bidding for property in close proximity to the US nation's capital, land prices in Washington, DC are high. To cover the cost of acquiring a building plot in that market, property developers had to charge steep rents. And that generally meant constructing multi-storey office buildings or apartments. Yet during the early 1990s, developers began constructing scores of golf driving ranges. A typical driving range might attract a few dozen customers each evening. Each would pay a few dollars for the privilege of driving golf balls into the night sky, but the total take in any given month would be too small to cover even the interest on the loan required to buy the land. Why did developers use their land in this way?

Washington area developers constructed new office buildings and apartments at a rapid rate throughout the late 1980s. Home prices and office rents had been rising rapidly, and developers had been busily acquiring undeveloped building lots in the expectation of further increases. The upshot was that when the national economic recession began in 1991, the Washington property market was substantially overbuilt. Vacancy rates soared and rents plummeted. Any developer who constructed a new office building or block of flats during that period could expect it to sit empty for a while.



The best use of a resource is not always a profitable one.
DRAWING BY MICK STEVENS

Instead of building on their undeveloped land, developers could sell it at a lower price or hold on to it until the market recovered. Those who followed the latter strategy had an obvious incentive to put their land to economic use during the interim. And for that purpose, a golf driving range is almost ideal. All it requires is a stock of used golf balls, a trailer from which to dispense them and a buggy to gather them up. These

investments were minimal and would be easy to liquidate when the property market recovered.

How could the meagre income from a driving range justify the opportunity cost of holding on to land purchased at such great expense? Clearly the developers never would have purchased the land when they did had they anticipated the imminent downturn. But given that they already owned the land and planned to sit out the downturn, their challenge was to make the best possible use of it in the interim. To make economic sense under these circumstances, a driving range need not generate enough revenue to cover the opportunity cost of the land on which it is built. Merely by generating more revenue than the marginal cost of operating them, the driving ranges made developers better off than if they had let their land sit idle.

THE FINAL EXAMPLES in this chapter discuss phenomena whose explanations require attention to both sides of the market, supply and demand.

Why are brown eggs more expensive than white ones in some countries? (*Jonathan Chang*)

In US supermarkets, jumbo eggs sell for \$3.09 a dozen if their shells are white but for \$3.79 a dozen if their shells are brown. According to the US Egg Nutrition Center in Washington, DC, neither the taste of an egg nor its nutritional quality depends on the colour of its shell. What explains this price difference?

It is tempting to say that brown eggs are more expensive because consumers prefer their looks and are willing to pay extra for them. But that observation does not constitute a satisfactory explanation because it seems to imply that sellers of white eggs are leaving cash on the table. If they could earn higher profits by selling brown eggs, why do they continue selling white ones?

A plausible answer is that brown eggs are more costly to produce than white ones. The colour of an egg depends on the breed of hen that lays it. White Leghorn hens, for example, lay white eggs, and Rhode Island Red hens lay brown ones. Brown hens tend to be larger than white ones, and since a hen's daily calorie requirement depends on its size, producing brown eggs costs more. But to explain why they sell for more, an important condition must also be present on the demand side. Unless some consumers prefer the looks of brown eggs and are willing to pay more for them, they will not be offered for sale.

Why is the average new house built in the UK so much smaller than the average new house in Australia and the US?

In 2005, the median new detached single family house in the US had 229 square metres of living space, roughly the same as in Australia, and more than three times as much as in the UK. Why are houses in the UK so much smaller?

The demand for housing increases with income. But median household income in the US is only about 20 per cent higher than in the UK, not enough to explain the observed difference in house size.

Income inequality is also higher in the US than in the UK, which could also contribute to the median difference in house size. A link between income inequality and average house size exists because the housing of the wealthy sets the frame of reference that defines adequate housing for the near wealthy, whose housing in turn sets the standard for others just below them, and so on down the income ladder. In the US, the income of the 90th percentile earner is 5.7 times that of the 10th percentile earner, as compared with a ratio of only 4.6 in the UK. Thus the median family in the US may build bigger in part because of the higher standard set by top earners there.

But these explanations can't be the whole story. Thus median household income in Australia is actually 5 per cent lower than in the UK, yet their houses are just as big as the Americans'. Income inequality is also slightly lower in Australia than in the UK. So some other factor must also be implicated in the observed differences in house size.

A plausible candidate is population density. Houses occupy land, and the fewer people there are per square kilometer of land, the cheaper land will be. Of the three countries, Australia has by far the lowest population density, only 2.6 persons per square kilometer. The United States has 31 persons per square kilometer, but that figure overstates the difference in density between it and Australia because a larger proportion of Australia is desert. In striking contrast, the UK has 246 persons per square kilometer. The resulting higher land prices in the UK lead rational families to economise by building smaller houses.

Why do film processing stores give you a second set of prints for free? (Laura Sandoval)

When you have a roll of film developed, many stores offer a second set of prints at no additional charge. Yet on any given roll of film, most pictures are not worth duplicating. So why do stores offer free duplicates of the duds, rather than offer the first set of prints at half price?

As noted earlier, most film is now processed automatically. The employee merely loads the roll of negatives and the machine does the rest. To generate a second set of prints, the employee presses a button. No additional labour time is required. The paper and chemicals needed for duplicate prints add to costs, but only slightly. So making two sets of prints costs only a little more.

On the buyer's side, even though most pictures on any given roll are duds, there are inevitably a few that are good enough to

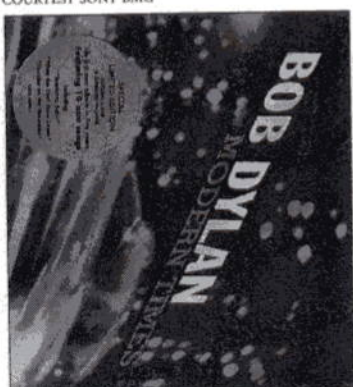
send to family or friends. Customers who get a single set of prints must identify the negatives of the pictures they want duplicated and then make an extra trip to the photo store. Duplicates made in this fashion require more care and attention from the equipment operator, so stores must charge high prices to cover their costs.

Stores that offer a second set of prints for free are thus offering a valuable service to their customers at only a minuscule increase in costs. Any store that failed to extend this offer would surely lose many of its customers to competitors.

Why do the most popular books and CDs sell for less than the least popular books and CDs, whereas we observe the opposite pattern for cinema ticket prices? (Ed Varga)

The list price of Bob Dylan's CD *Modern Times* is £16.99, but it is available on amazon.co.uk for only £6.98, a large discount. By contrast, recordings by less popular artists are sold at much

£6.98



£13.99



Are recording artists exempt from the laws of supply and demand?

smaller discounts. For example, Paris Combo's *Motifs* is sold on amazon.co.uk for £13.99, at a negligible discount. The pattern is similar for books. Borders bookstore, for example, offers a 25 per cent discount on best-selling books but charges the publisher's suggested retail price for most other titles.

The pattern is the reverse for cinema tickets. Although the stated admission price is typically the same for all films shown at a given time at a given cinema, cinema owners are much less likely to offer discount tickets for hit films than for other films. Why are cinema operators, but not book and CD sellers, exploiting the fact that buyers are willing to pay more for more popular products?

Each book, film and CD is unique. Because rival sellers cannot offer perfect substitutes for these products, the markets for them are not perfectly competitive. Even so, the general pattern in less than perfectly competitive markets is for prices to be higher for products and services that buyers value most highly. This, as noted, is the pattern we observe with films.

A plausible explanation for why books and CDs depart from this pattern begins with the observation that the cost conditions for sellers of these products differ sharply from those of cinema owners. For cinemas, the scarce resource that determines how prices are set is not films but seats. Once a cinema's seats are filled, it is impossible to serve an additional customer at any price. So cinema owners have a strong incentive not to offer discounts on films that can fill the seats at the regular admission price. Book and music sellers, by contrast, are unlikely to have to turn away customers if they offer discounts on popular items. Most of the time, they can anticipate which items will be most popular and can accommodate demands by keeping more copies in stock. And since these items turn over rapidly, the per-copy cost of the shelf space required to keep them in stock is very low. Less popular books and CDs, which might sell only a copy or

two every few months, generate less revenue for the same shelf space and thus are more costly to keep in stock.

Virtually all retailers stock the most popular books and CDs (since they know there will be a brisk market for them), but there is less overlap in the lists of more obscure entries kept in stock at different stores. And this means retail outlets will face more competitive pressure from one another on the most popular entries. Someone who doesn't like one store's price for a new Dylan CD can buy it at any number of other stores. But few other stores are likely to have the latest Paris Combo CD in stock. Someone who wants that disc right away has little alternative but to pay the supplier's price.

The most successful book and music stores steer their customers to promising but obscure new entries that might otherwise escape their attention. It is thus the least popular titles that are most responsible for the cost of recruiting the knowledgeable sales staff required to make these connections. Popular items sell for bigger discounts in part because they are so much cheaper to sell. So the next time you sit down to listen to that great new Paris Combo CD, just remember that you paid more for it than for the popular CDs you can buy at Woolworths because your music store bore the additional expense of recruiting a salesperson knowledgeable enough to know you'd probably like it.

Yet another impetus for stores to discount best-selling books and CDs is that the practice attracts additional customers to stores, who are likely to buy other items.